Problem 15.4.3

Find the Maclaurin series and its radius of convergence.

$$\sin 2z^2$$

Solution

$$\sin(w) = \sum_{n=1}^{\infty} (-1)^n \frac{w^{2n+1}}{(2n+1)!}$$

$$\sin 2z^2 = \sum_{n=1}^{\infty} (-1)^n \frac{(2z^2)^{2n+1}}{(2n+1)!} = \boxed{\sum_{n=1}^{\infty} (-1)^n \frac{2^{2n+1} z^{4n+2}}{(2n+1)!}}$$

$$\begin{aligned} L &= \lim_{n\to\infty} \left| \frac{(-1)^{n+1}}{(-1)^n} \cdot \frac{2^{2n+3}}{2^{2n+1}} \cdot \frac{(2n+1)!}{(2n+3)!} \right| \\ &= \lim_{n\to\infty} \left| -1 \cdot 4 \cdot \frac{1}{(2n+3)(2n+2)} \right| \\ &= 0 \\ \therefore R &= \boxed{\infty} \end{aligned}$$

Problem 15.4.16. Inverse sine.

Developing $1/\sqrt{1-z^2}$ and integrating, show that

$$\sin^{-1} z = z + \left(\frac{1}{2}\right)\frac{z^3}{3} + \left(\frac{1\cdot 3}{2\cdot 4}\right)\frac{z^5}{5} + \left(\frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\right)\frac{z^7}{7} + \cdots (|z|<1)$$

Solution

$$f(z) = \sin^{-1} z$$

$$f'(z) = \left(1-z^2\right)^{-1/2}$$

$$(1+w)^{-m} = \sum_{n=0}^{\infty}\binom{-m}{n} w^n = 1 - mw + \frac{m(m+1)}{2!}w^2 - \frac{m(m+1)(m+2)}{3!}w^3 + \cdots$$

$$\begin{aligned}
f'(z) &= \sum_{n=0}^{\infty}\binom{\frac{1}{2}}{n}\left(-z^2\right)^n \\
&= 1 - \frac{1}{2}\left(-z^2\right) + \frac{\frac{1}{2}(\frac{1}{2}+1)}{2!}\left(-z^2\right)^2 - \frac{\frac{1}{2}(\frac{1}{2}+1)(\frac{1}{2}+2)}{3!}\left(-z^2\right)^3 + \cdots \\
&= 1 + \frac{1}{2}z^2 + \frac{\frac{1}{2}(\frac{3}{2})}{2}z^4 + \frac{\frac{1}{2}(\frac{3}{2})(\frac{5}{2})}{6}z^6 + \cdots \\
&= 1 + \left(\frac{1}{2}\right)z^2 + \left(\frac{1\cdot 3}{2\cdot 4}\right)z^4 + \left(\frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\right)z^6 + \cdots
\end{aligned}$$

Integrating termwise

$$f(z) = z + \left(\frac{1}{2}\right)\frac{z^3}{3} + \left(\frac{1\cdot 3}{2\cdot 4}\right)\frac{z^5}{5} + \left(\frac{1\cdot 3\cdot 5}{2\cdot 4\cdot 6}\right)\frac{z^7}{7} + \cdots$$

This converges on $|z|<1$ due to the geometric series f'

Problem 15.4.19

Find the Taylor series with center $z_0 = i$ and its radius of convergence.

$$\frac{1}{1-z}$$

Solution

$$\begin{aligned}
f(z) &= \frac{1}{1-z} \\
&= \frac{1}{1-i-(z-i)} \\
&= \frac{1}{(1-i)\left(1-\frac{z-i}{1-i}\right)} \\
&= \frac{1}{1-i}\sum_{n=0}^{\infty}\left(\frac{z-i}{1-i}\right)^n \quad \text{by 15.4(11)}
\end{aligned}$$

This series converges on

$$\begin{aligned}
\left|\frac{z-i}{1-i}\right| &< 1 \\
|z-i| &< |1-i| \\
|z-i| &< \sqrt{2} \\
\therefore R &= \boxed{\sqrt{2}}
\end{aligned}$$

Problem 15.4.21

Find the Taylor series with center $z_0 = \frac{\pi}{2}$ and its radius of convergence.

$$\sin z$$

Solution

$$\begin{aligned}
f(z) &= \sin z & f(z_0) &= 1 \\
f'(z) &= \cos z & f'(z_0) &= 0 \\
f''(z) &= -\sin z & f''(z_0) &= -1 \\
f^{(3)}(z) &= -\cos z & f^{(3)}(z_0) &= 0 \\
f^{(4)}(z) &= \sin z & f^{(4)}(z_0) &= 1
\end{aligned}$$

$$\begin{aligned}
f(z) &= \sum_{n=0}^{\infty} a_n (z - z_0)^n & a_n &= \frac{1}{n!} f^{(n)}(z_0) \\
&= 1 + 0 - \frac{1}{2!}(z - \frac{\pi}{2})^2 + 0 + \frac{1}{4!}(z - \frac{\pi}{2})^4 + \cdots \\
&= 1 - \frac{1}{2!}(z - \frac{\pi}{2})^2 + \frac{1}{4!}(z - \frac{\pi}{2})^4 - \cdots \\
&= \boxed{\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!}(z - \frac{\pi}{2})^{2n}}
\end{aligned}$$

$$\begin{aligned}
L &= \lim_{n\to\infty} \left| \frac{a_{n+1}}{a_n} \right| \\
L &= \lim_{n\to\infty} \left| \frac{(-1)^{n+1}}{(2n+2)!} \cdot \frac{(2n)!}{(-1)^n} \right| \\
L &= \lim_{n\to\infty} \left| \frac{-1}{(2n+2)(2n+1)} \right| \\
L &= 0 \\
\therefore R &= \boxed{\infty}
\end{aligned}$$

Problem 15.4.23

Find the Taylor series with center $z_0 = i$ and its radius of convergence.

$$\frac{1}{(z+i)^2}$$

Solution